Software Requirements Specification

Starlink Wi-Fi Subscription Portal

Captive-portal access control with Fapshi mobile money payments on MikroTik

Document	Software Requirements Specification (SRS)
Project	Starlink Wi-Fi Subscription Portal (working title)
Version	1.0 (Draft)
Date	1 October 2026
Prepared by	Mbuh Favour, Project Owner
Status	Draft for review and sign-off

Draft for review. Structure follows IEEE 830 and ISO/IEC/IEEE 29148 conventions.

Document Control

Revision history

Version	Date	Author	Description
1.0	1 October 2026	Project Owner	First complete draft of the SRS.

Document conventions

Requirements are written with the word **shall** for mandatory behaviour and **should** for recommended behaviour. Each requirement has a unique identifier such as FR-PY-03 (functional requirement, payment group, item 3). Priorities follow the MoSCoW scheme.

Priority	Meaning
Must	Essential for the first release. The system is not acceptable without it.
Should	Important and expected, but the system can launch without it if time is short.
Could	Desirable improvement that is included only if time and budget allow.
TBD	Value or decision not yet fixed. Listed in section 2.7 (Open decisions).

Table of Contents

1. Introduction 6

1.1 Purpose . . . 6
1.2 Scope . . . 6
1.3 Intended audience . . . 6
1.4 Definitions, acronyms and abbreviations . . . 6
1.5 References . . . 7
1.6 Document overview . . . 7

2. Overall Description 8

2.1 Product perspective . . . 8
2.2 Product functions . . . 8
2.3 User classes and characteristics . . . 8
2.4 Operating environment . . . 9
2.5 Design and implementation constraints . . . 9
2.6 Assumptions and dependencies . . . 9
2.7 Open decisions . . . 9

3. System Architecture 11

3.1 Component overview . . . 11
3.2 Subscription flow . . . 11
3.3 Administrator flows . . . 11
3.4 Subscription lifecycle . . . 12

4. Functional Requirements 13

4.1 Captive portal and login . . . 13
4.2 Plans . . . 13
4.3 Payment . . . 13
4.4 Access provisioning . . . 14
4.5 Device binding . . . 14
4.6 Anti-tethering and fair use . . . 14
4.7 Tenant status and help . . . 15
4.8 Administration portal . . . 15
4.9 Vouchers for cash payments . . . 16
4.10 Outage and failure handling . . . 16
4.11 Logging and reporting . . . 16

5. External Interface Requirements 17

5.1 User interfaces . . . 17
5.2 Hardware interfaces . . . 17
5.3 Software interfaces . . . 17
5.4 Communications interfaces . . . 18

6. Data Requirements 19

6.1 Main entities . . . 19
6.2 Reference schema . . . 19
6.3 Data protection and retention . . . 20

7. Network and Router Configuration Requirements 21

8. Non-Functional Requirements 22

8.1 Performance and capacity . . . 22
8.2 Security . . . 22
8.3 Reliability and availability . . . 22
8.4 Usability . . . 23
8.5 Maintainability and portability . . . 23
8.6 Legal and commercial . . . 23

9. Use Cases 24

9.1 UC-01 Tenant buys a plan . . . 24
9.2 UC-02 Tenant checks status . . . 24
9.3 UC-03 Owner approves a tenant's laptop . . . 24
9.4 UC-04 Owner re-binds a subscription . . . 24
9.5 UC-05 Owner issues a cash voucher . . . 25
9.6 UC-06 Starlink outage . . . 25

10. Technology Stack and Project Structure 26

11. Risk Analysis 26

12. Testing and Acceptance Criteria 28

12.1 Test approach . . . 28
12.2 Acceptance criteria . . . 28

13. Deployment and Operations 28

13.1 Deployment steps . . . 28
13.2 Ongoing operations . . . 29

14. Future Enhancements 29

15. Appendix A: Traceability Summary 30

16. Appendix B: Items to Confirm Before Development 30

1. Introduction

1.1 Purpose

This document specifies the requirements for the Starlink Wi-Fi Subscription Portal. The system lets the owner of a Starlink internet connection sell time-limited Wi-Fi access to tenants. A tenant who joins the Wi-Fi without having paid is shown a sign-in page where they can buy a plan using mobile money. After payment is confirmed, that tenant's device is given internet access for the purchased period, and nothing else needs to be done by hand.

The document is intended to be the single agreed description of what the system must do, so that design, development, installation and testing can all be checked against it.

1.2 Scope

In scope

- A captive portal on a MikroTik router that blocks internet access until a valid subscription exists.
- A tenant-facing website (plans, payment, status, help) built with plain HTML, CSS and JavaScript.
- A backend written in raw PHP (no framework) that talks to the Fapshi payment API and to the MikroTik router.
- Binding each subscription to a single device (MAC address) so one payment covers one device.
- Measures against tethering and hotspot sharing from a paid phone.
- An administration portal for the owner: device approval (for example tenant laptops), revoke, extend, re-bind, cash vouchers, payment records and audit log.
- Handling of Starlink outages, payment failures and router unavailability.

Out of scope (first release)

- Data-volume quotas and usage-based billing.
- Multiple sites or multiple routers managed from one portal.
- Native mobile applications.
- Card payments or payment providers other than Fapshi.
- Automatic recurring billing or auto-renewal.

1.3 Intended audience

Reader	Use of this document
Owner (project sponsor)	Confirms that the described behaviour, plans and controls match the business need.
Developer	Implements the portal, PHP backend and admin portal from the requirements, data model and interfaces.
Network technician	Configures the Starlink bypass, MikroTik hotspot, firewall rules and API access from sections 5 and 7.
Tester	Derives test cases from the functional requirements, use cases and acceptance criteria.

1.4 Definitions, acronyms and abbreviations

Term	Definition
Captive portal	A page that unauthenticated users are automatically redirected to before they can use the internet.

Term	Definition
Hotspot (MikroTik)	The RouterOS feature that provides captive-portal login, user accounts, time limits and speed limits.
Walled garden	A list of addresses that unauthenticated devices are allowed to reach, for example the portal server.
MAC address	The hardware identifier of a device's network interface, used here to bind a subscription to one device.
Private (random) MAC	A per-network MAC address generated by modern phones for privacy. It is stable for a network unless the user resets it.
TTL	Time To Live, a counter in each IP packet that is reduced by one at every router hop. Tethered traffic shows a lower value.
Tethering	Sharing one device's internet connection with other devices using Wi-Fi hotspot, USB or Bluetooth.
Fapshi	Cameroonian payment service used to collect MTN Mobile Money and Orange Money payments.
MoMo / OM	MTN Mobile Money / Orange Money.
USSD	A mobile network service used to deliver the payment confirmation prompt to the tenant's phone.
CGNAT	Carrier-grade NAT. Starlink connections sit behind it, so the network cannot be reached directly from the internet.
Webhook	An HTTP request sent by Fapshi to the portal server when a payment changes state.
Voucher	A single-use access code generated by the owner for cash payments.
XAF (FCFA)	Central African CFA franc, the currency of all prices in this system.
SRS	Software Requirements Specification.

1.5 References

- Fapshi developer documentation: payment initiation, transaction status and webhook behaviour (exact endpoints and fields to be confirmed against the current documentation before implementation).
- MikroTik RouterOS documentation: Hotspot, User Profiles, Walled Garden, Firewall and the RouterOS API (classic API and REST API in RouterOS 7).
- IEEE Std 830-1998 and ISO/IEC/IEEE 29148:2018: guidance on requirements specifications.
- Starlink documentation on bypass mode and Ethernet adapters for the specific dish model in use.

1.6 Document overview

Section 2 describes the product as a whole. Section 3 describes the architecture and key flows. Section 4 lists the functional requirements. Section 5 covers external interfaces, section 6 the data model, section 7 the router configuration, and section 8 the non-functional requirements. Sections 9 to 13 contain use cases, the technology stack, risks, acceptance criteria and deployment notes. Section 14 lists possible future work.

2. Overall Description

2.1 Product perspective

The system is a small, self-contained solution that sits between the Starlink internet source and the tenants. It has three cooperating parts: the MikroTik router (which enforces who is online), the portal server (which sells plans and talks to Fapshi) and the tenants' and owner's browsers. The Starlink router is placed in bypass mode so that the MikroTik becomes the only gateway for tenant devices.

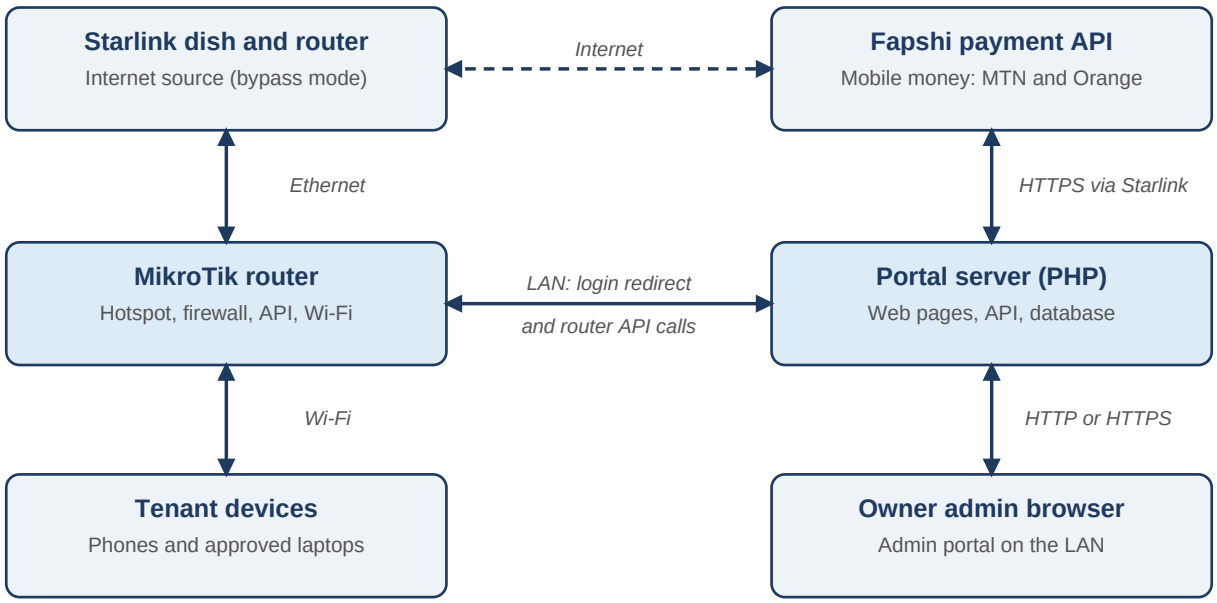


Figure 1. System context and main connections.

2.2 Product functions

- Intercept unpaid devices and show a sign-in page with a Buy a plan option.
- Offer several plans with different durations and speeds.
- Collect a mobile money payment through Fapshi and confirm it server-side.
- Create a time-limited hotspot account bound to the paying device, and log the device in automatically.
- Allow only the paying device to use the subscription.
- Block tethering and apply per-user speed and connection limits.
- Let tenants see their plan, expiry and payment status.
- Give the owner an admin portal to approve extra devices (for example laptops), revoke or extend access, re-bind a changed device, issue cash vouchers and review payments.
- Warn tenants and disable purchases when the Starlink connection is down.

2.3 User classes and characteristics

User class	Description	Technical level
Tenant	Resident who buys Wi-Fi access. Most tenants are students who use a single phone, on low-end to mid-range Android and iPhone models. Some also have a laptop.	Basic. Needs a very simple, mobile-first flow.

User class	Description	Technical level
Owner / administrator	Person who owns the Starlink connection and the router. Approves extra devices, handles cash payments and resolves problems.	Moderate. Comfortable with web forms, not necessarily with router settings.
Technician / developer	Installs and maintains the router configuration and the portal server.	Advanced.

2.4 Operating environment

Component	Environment
Internet source	Starlink dish with its router in bypass mode. An Ethernet adapter may be required for some dish models.
Gateway	MikroTik router running RouterOS 7 (for example hAP ac2 or hAP ax2), with the Hotspot feature and API service enabled.
Portal server	Always-on small computer on the LAN (laptop, mini PC or Raspberry Pi) running Linux or Windows, a web server (Apache or Nginx) and PHP 8.x. Recommended deployment, to be confirmed in section 2.7.
Database	SQLite or MySQL/MariaDB on the portal server.
Tenant devices	Any current mobile or desktop browser that supports captive-portal detection.
Power	Mains power with a UPS or battery backup for the router and portal server.

2.5 Design and implementation constraints

- The backend shall be written in raw PHP without a framework, because it is the language the developer is familiar with.
- The tenant frontend shall use plain HTML, CSS and JavaScript only, with about four pages.
- Payments shall use Fapshi. The Fapshi API credentials shall never be exposed to browsers.
- The router platform is MikroTik. Access control is performed by the MikroTik Hotspot, not by the website.
- Hosting cost shall be kept as low as possible, preferably free.
- The solution shall work on a Starlink connection behind CGNAT, so the router cannot be reached from the internet.

2.6 Assumptions and dependencies

- The Starlink service is active and the dish has a clear view of the sky.
- Fapshi remains available and its API stays compatible with the integration described here.
- Tenants have an active MTN or Orange mobile money account and mobile network coverage to receive the payment prompt (USSD does not need mobile data).
- Tenants' devices keep a stable MAC address for the Wi-Fi network while a subscription is active.
- The portal server has a stable local IP address and can reach the internet through the MikroTik.
- The router clock is correct (time synchronisation is enabled), because expiry depends on it.

2.7 Open decisions

ID	Decision needed	Current default / proposal
OD-1	When does the subscription clock start: at payment, at first login, or counted only while online?	Proposed: calendar validity starting at first login, with a maximum activation window after payment.
OD-2	Plan names, durations, speeds and prices (XAF).	TBD by the owner. See section 4.2.

ID	Decision needed	Current default / proposal
OD-3	Where the portal server runs: local device on the LAN or a free cloud host.	Recommended: local device (direct router access, works when the internet is down).
OD-4	How Fapshi payments are collected: hosted checkout page or direct charge to the phone number.	Preferred: direct charge, so tenants never leave the portal. Confirm with Fapshi documentation.
OD-5	How payment confirmation is received: webhook, polling, or both.	Proposed: polling as the default, with webhook if a tunnel is added.
OD-6	Whether tenants may buy an extra-device add-on themselves.	Could-have. Default: all extra devices go through the owner.
OD-7	Router model and RouterOS version.	TBD. A RouterOS 7 device is assumed.
OD-8	Data retention period for payment and device records.	Proposed: 12 months, then anonymise or delete.
OD-9	Interface languages.	Proposed: English and French.

3. System Architecture

3.1 Component overview

Component	Responsibility
Starlink dish and router	Provides the internet connection. The router runs in bypass mode and does not control tenants.
MikroTik router	Wi-Fi and wired LAN, DHCP, NAT, Hotspot captive portal, per-user profiles, expiry, walled garden, TTL and connection limiting, and the API used by the portal server.
Tenant frontend	Static HTML, CSS and JavaScript pages served by the portal server: plans, payment, status and help.
PHP backend	Creates orders, calls Fapshi, confirms payments, provisions and removes hotspot users on the router, serves status data and runs scheduled reconciliation.
Admin portal	Password-protected PHP pages for the owner.
Database	Stores plans, orders, subscriptions, devices, vouchers, admins and audit entries.
Fapshi	Collects mobile money payments and reports their status.

3.2 Subscription flow

Step	Actor	Action
1	Tenant	Connects a device to the Wi-Fi. The MikroTik detects it has no active subscription.
2	MikroTik	Intercepts web traffic and shows the hotspot login page. The page offers existing-code login and a Buy a plan button.
3	Tenant	Taps Buy a plan. The router redirects to the portal and passes the device MAC address and IP address.
4	Portal	Shows the plans. The tenant selects one and enters a mobile money phone number.
5	Backend	Creates an order with a unique reference and status Pending, then asks Fapshi to start the payment.
6	Tenant	Approves the payment on the phone with the mobile money PIN.
7	Backend	Receives the Fapshi notification or polls Fapshi, then verifies the transaction status directly with Fapshi.
8	Backend	If the payment is successful and has not been processed before, creates a hotspot user named after the device MAC address with the plan's limits.
9	MikroTik	Logs the device in automatically through MAC login. Internet access begins.
10	Portal	Status page shows the plan and expiry. Router enforces expiry and removes access at the end.

3.3 Administrator flows

- **Approve a device:** the tenant connects a laptop, the owner opens the admin portal, picks the device from the list of connected but not logged-in devices, selects tenant and plan, and approves it.
- **Cash voucher:** the owner generates a single-use code, hands it to the tenant, who enters it on the login page. The code binds to that tenant's device.
- **Re-bind:** when a tenant's device MAC changes, the owner moves the active subscription to the new MAC address.
- **Revoke or extend:** the owner stops a subscription immediately or adds time to it.

3.4 Subscription lifecycle

State	Meaning	Possible next states
Pending	Order created, waiting for Fapshi confirmation.	Paid, Failed, Expired order
Paid	Payment confirmed, router user not yet created (for example router unreachable).	Active, Paid (retry)
Active	Hotspot user exists and the time period is running or available.	Expired, Revoked, Active (extended or re-bound)
Expired	Purchased period ended. Router user removed or disabled.	None (tenant buys again)
Revoked	Stopped by the owner.	None (tenant buys again or owner re-approves)
Failed	Payment declined, cancelled or timed out.	None (tenant may retry with a new order)

4. Functional Requirements

4.1 Captive portal and login

ID	Requirement	Priority
FR-CP-01	The router shall intercept every unauthenticated device and redirect its web traffic to the hotspot login page.	Must
FR-CP-02	The login page shall present a Buy a plan button and a field to enter an existing voucher code.	Must
FR-CP-03	The redirect to the portal shall carry the device MAC address and IP address as parameters.	Must
FR-CP-04	The walled garden shall allow unauthenticated devices to reach the portal server and nothing else, unless hosted checkout is used, in which case the required Fapshi domains shall also be allowed.	Must
FR-CP-05	The login page and portal pages shall be light enough to load quickly on low-end phones.	Must
FR-CP-06	The login page shall work with the automatic captive-portal pop-up on Android, iOS, Windows and macOS, and shall also be reachable manually by opening a plain HTTP address.	Should
FR-CP-07	The login page shall show a service-down notice when the Starlink connection is unavailable (see FR-OU-02).	Should

4.2 Plans

ID	Requirement	Priority
FR-PL-01	The system shall support multiple plans, each with a name, duration, price in XAF, speed profile and number of allowed devices.	Must
FR-PL-02	The plans page shall list all active plans with price and duration, using clear language.	Must
FR-PL-03	The owner shall be able to create, edit, disable and reorder plans from the admin portal without editing code.	Should
FR-PL-04	Changing a plan shall not alter subscriptions already purchased.	Must
FR-PL-05	A plan shall define whether its duration is counted from payment, from first login, or as online time only (see OD-1).	Should

Plan (example)	Duration	Speed profile	Devices	Price (XAF)
Daily	24 hours	TBD	1	TBD
Weekly	7 days	TBD	1	TBD
Monthly	30 days	TBD	1	TBD
Extra device (optional)	Matches parent plan	Same as parent	+1	TBD

Table 4.2. Illustrative plan structure. Names and prices are decided by the owner (OD-2).

4.3 Payment

ID	Requirement	Priority
FR-PY-01	The tenant shall enter a mobile money phone number, which the system shall validate against the Cameroonian format before sending it to Fapshi.	Must
FR-PY-02	Before contacting Fapshi, the backend shall create an order record with a unique reference, the plan, the device MAC address and the status Pending.	Must
FR-PY-03	The backend shall start the payment with Fapshi using credentials stored on the server only.	Must

ID	Requirement	Priority
FR-PY-04	The backend shall confirm every payment by querying Fapshi for the transaction status. A webhook or a browser redirect alone shall never grant access.	Must
FR-PY-05	The system shall support confirmation by polling Fapshi, and by webhook where a public URL is available.	Must
FR-PY-06	Each Fapshi transaction identifier shall be stored as unique, so repeated notifications can never create two subscriptions.	Must
FR-PY-07	The portal shall show the tenant a clear pending screen while waiting, and a clear success or failure message afterwards.	Must
FR-PY-08	Orders that remain Pending beyond a configurable timeout (default 10 minutes) shall be marked Expired order, after a final status check with Fapshi.	Must
FR-PY-09	A tenant whose payment failed or was cancelled shall be able to retry without contacting the owner.	Should
FR-PY-10	The system shall never ask for, receive or store mobile money PINs.	Must

4.4 Access provisioning

ID	Requirement	Priority
FR-AP-01	After a confirmed payment, the backend shall create a hotspot user on the MikroTik named after the paying device's MAC address, with the plan's profile and time limit.	Must
FR-AP-02	The device shall be logged in automatically through MAC login, or shown a single-tap login button if automatic login is not possible.	Must
FR-AP-03	Provisioning shall complete within 10 seconds of payment confirmation under normal conditions.	Should
FR-AP-04	If the router API is unreachable, the order shall be marked Paid, retried automatically at short intervals, and flagged on the admin dashboard.	Must
FR-AP-05	The router shall enforce expiry by itself, so access ends on time even if the portal server is off.	Must
FR-AP-06	Expired hotspot users shall be removed or disabled by the router, and the portal database shall be updated by a scheduled job.	Should
FR-AP-07	A tenant who buys a second plan while one is active shall have the new time added to the existing subscription.	Should

4.5 Device binding

ID	Requirement	Priority
FR-DB-01	One paid subscription shall authorise exactly one device, identified by its MAC address.	Must
FR-DB-02	A second device attempting to use the same subscription shall be shown the login page and shall have to pay or be approved separately.	Must
FR-DB-03	Each hotspot user shall be limited to one simultaneous session.	Must
FR-DB-04	If a tenant's MAC address changes (for example after resetting the Wi-Fi network on the phone), the portal shall show a clear message to contact the owner.	Should
FR-DB-05	The owner shall be able to re-bind an active subscription to a new MAC address.	Must
FR-DB-06	The system shall store the device MAC address with each order and subscription for audit purposes.	Must

4.6 Anti-tethering and fair use

ID	Requirement	Priority
FR-AT-01	The router shall apply a TTL filter on the hotspot network that accepts the normal TTL values 64 and 128 and drops packets with other values, which indicate traffic forwarded through another device.	Should
FR-AT-02	The filter shall also cover IPv6 traffic, or IPv6 shall be disabled on the hotspot network.	Should
FR-AT-03	Every plan profile shall include a per-user speed limit.	Must
FR-AT-04	Every plan profile shall include a per-user connection limit.	Should
FR-AT-05	Directly connected laptops approved by the owner shall not be affected by the filter.	Must
FR-AT-06	The owner shall be able to switch the filter on or off on the router without changing the portal software.	Should
FR-AT-07	The system shall be documented as providing best-effort control only: a VPN or a modified TTL value can bypass the filter.	Must

4.7 Tenant status and help

ID	Requirement	Priority
FR-ST-01	A logged-in tenant shall be able to see the current plan, expiry date and time, and remaining time.	Must
FR-ST-02	The status page shall identify the tenant only through the device that is connected, not through a typed phone number.	Should
FR-ST-03	The help page shall explain how to pay, what to do if payment succeeded but there is no access, the one-device rule, and how to contact the owner.	Must
FR-ST-04	Tenants shall be able to see their last few payments and their reference numbers.	Could
FR-ST-05	Pages shall be available in English and French.	Should

4.8 Administration portal

ID	Requirement	Priority
FR-AD-01	Admin pages shall require login with a username and password. Passwords shall be stored as salted hashes.	Must
FR-AD-02	The portal shall list devices that are connected to the router but not logged in, showing MAC address, IP address and host name.	Must
FR-AD-03	The owner shall be able to approve a device from that list by choosing a tenant name and a plan. The system shall then create a MAC-bound hotspot user.	Must
FR-AD-04	The owner shall be able to revoke a subscription immediately, which also disconnects the active session.	Must
FR-AD-05	The owner shall be able to extend a subscription by a chosen amount of time.	Should
FR-AD-06	The owner shall be able to re-bind a subscription to another MAC address.	Must
FR-AD-07	The dashboard shall show active subscriptions, today's revenue, pending or stuck payments and provisioning errors.	Should
FR-AD-08	The owner shall be able to search payments by phone number, reference, MAC address and date.	Should
FR-AD-09	The owner shall be able to manage plans (see FR-PL-03).	Should
FR-AD-10	Every administrative action shall be recorded in the audit log with time, administrator and details.	Must

ID	Requirement	Priority
FR-AD-11	The admin portal shall not be reachable by unauthenticated tenants through the walled garden, and shall accept connections only from the local network or configured addresses.	Must

4.9 Vouchers for cash payments

ID	Requirement	Priority
FR-VC-01	The owner shall be able to generate one or more vouchers for a chosen plan.	Should
FR-VC-02	A voucher shall be single-use. When redeemed on the login page, it shall bind the subscription to the redeeming device.	Should
FR-VC-03	Vouchers shall have an expiry date after which they cannot be redeemed.	Could
FR-VC-04	Vouchers shall be generated with enough randomness to prevent guessing, and failed redemption attempts shall be rate limited.	Should

4.10 Outage and failure handling

ID	Requirement	Priority
FR-OU-01	The portal server shall regularly check that the internet connection and the Fapshi API are reachable.	Should
FR-OU-02	When the connection is down, the portal shall show a service-temporarily-down notice and disable the Pay button.	Should
FR-OU-03	If a payment was started just before an outage, the system shall reconcile it with Fapshi when the connection returns and provision access if the payment succeeded.	Must
FR-OU-04	Cash vouchers and owner approvals shall keep working while the internet is down, as long as the portal server and router are running.	Should
FR-OU-05	The system shall continue to enforce existing subscriptions on the router even if the portal server is off.	Must

4.11 Logging and reporting

ID	Requirement	Priority
FR-LG-01	The system shall log payment events, Fapshi responses (without secrets), router API calls and errors.	Must
FR-LG-02	The system shall provide a reconciliation report that compares Fapshi transactions with subscriptions created.	Should
FR-LG-03	The owner shall be able to export payments and subscriptions as CSV for a chosen date range.	Could

5. External Interface Requirements

5.1 User interfaces

Tenant pages (plain HTML, CSS and JavaScript, mobile-first):

Page	Purpose	Key elements
Plans (index)	Show available plans and start a purchase.	Plan cards with price, duration and speed. Service-status banner. Language switch.
Checkout	Collect the phone number and show payment progress.	Phone number field, payment network hint, pending animation, success and failure messages with Retry.
Status	Show the current subscription.	Plan name, expiry, remaining time, device note, recent payments.
Help	Support and rules.	Step-by-step payment guide, one-device rule, no-access troubleshooting, owner contact details.

Admin pages (PHP, for the owner):

Page	Purpose
Login	Authenticate the owner.
Dashboard	Active subscriptions, revenue today, stuck payments, router and Fapshi status.
Devices	Connected and not-yet-logged-in devices with an Approve action.
Subscriptions	Search, revoke, extend, re-bind.
Payments	All orders and transactions with filters and CSV export.
Vouchers	Generate and list vouchers.
Plans	Create and edit plans.
Audit log	History of administrative actions.

Interface principles

- Large touch targets, readable font sizes and high contrast.
- No dependence on large images, external fonts or third-party scripts, because unpaid devices can reach only the portal server.
- Plain language messages that tell the tenant exactly what to do next.

5.2 Hardware interfaces

Item	Interface
Starlink dish and router	Ethernet from the Starlink router (in bypass mode) to the MikroTik WAN port. An Ethernet adapter may be needed for some dish models.
MikroTik router	WAN port to Starlink, bridged LAN ports and Wi-Fi to tenants, LAN port to the portal server.
Portal server	Ethernet to the MikroTik LAN. Static or reserved IP address.
UPS or battery backup	Powers the router and portal server during power cuts so subscriptions keep running.

5.3 Software interfaces

5.3.1 Fapshi payment API

Aspect	Description
Direction	Backend to Fapshi (outbound HTTPS). Optionally Fapshi to backend for webhooks.
Operations used	Initiate a payment, query transaction status, receive payment-status notifications.
Data sent	Amount in XAF, phone number, order reference, optional description.
Data received	Transaction identifier, status (for example created, successful, failed, expired), amounts and timestamps.
Authentication	API user and API key held in a server-side configuration file outside the web root.
Notes	Exact endpoints, field names and status values must be confirmed against the current Fapshi documentation. The integration shall be isolated in one PHP module so changes affect only that module.

5.3.2 MikroTik RouterOS API

Operation	Purpose
Add hotspot user	Create the MAC-named user with profile, limits and comment (order reference).
Remove or disable hotspot user	Revoke or expire a subscription.
List hosts	Show connected devices that are not yet authorised (for the Devices page).
List active sessions	Show who is online and remove a session on revoke.
Read user and profile data	Reconcile router state with the database.
Check access	Connectivity test used for health monitoring.

The API shall be accessed by a dedicated low-privilege router account, using the encrypted API service or REST over HTTPS where possible, and restricted to the portal server's IP address.

5.4 Communications interfaces

- Tenant devices to portal server: HTTP or HTTPS over the LAN, allowed through the walled garden.
- Portal server to MikroTik: RouterOS API over the LAN, restricted by firewall to the portal server address.
- Portal server to Fapshi: outbound HTTPS through the Starlink connection.
- Optional: a free tunnel service to give the portal a public HTTPS address if Fapshi webhooks are used. Polling works without it.

6. Data Requirements

6.1 Main entities

Entity	Description	Key relationships
plans	Sellable plans with duration, price and router profile name.	One plan has many orders and subscriptions.
orders	A payment attempt with Fapshi reference, phone number, device MAC and status.	Belongs to a plan. May produce one subscription.
subscriptions	A granted access period bound to one MAC address.	Belongs to a plan and optionally an order or voucher.
vouchers	Cash-payment codes.	Belongs to a plan. May be redeemed into a subscription.
admins	Owner accounts for the admin portal.	Referenced by the audit log.
audit_log	Record of administrative and system actions.	References an admin where applicable.

6.2 Reference schema

The schema below is a starting point written in generic SQL. Column types may be adjusted for SQLite or MySQL. One statement or column is written per line for readability.

```
CREATE TABLE plans (
    id INTEGER PRIMARY KEY,
    name VARCHAR(60) NOT NULL,
    duration_minutes INTEGER NOT NULL,
    price_xaf INTEGER NOT NULL,
    router_profile VARCHAR(60) NOT NULL,
    max_devices INTEGER NOT NULL DEFAULT 1,
    is_active INTEGER NOT NULL DEFAULT 1,
    sort_order INTEGER NOT NULL DEFAULT 0
);
```

```
CREATE TABLE orders (
    id INTEGER PRIMARY KEY,
    reference VARCHAR(40) NOT NULL UNIQUE,
    plan_id INTEGER NOT NULL,
    phone VARCHAR(20) NOT NULL,
    device_mac VARCHAR(17) NOT NULL,
    amount_xaf INTEGER NOT NULL,
    fapshi_trans_id VARCHAR(64) UNIQUE,
    status VARCHAR(20) NOT NULL DEFAULT 'pending',
    created_at DATETIME NOT NULL,
    confirmed_at DATETIME,
    FOREIGN KEY (plan_id) REFERENCES plans (id)
);
```

```
CREATE TABLE subscriptions (
    id INTEGER PRIMARY KEY,
    order_id INTEGER,
    voucher_id INTEGER,
    plan_id INTEGER NOT NULL,
    device_mac VARCHAR(17) NOT NULL,
    tenant_name VARCHAR(80),
    status VARCHAR(20) NOT NULL DEFAULT 'active',
    starts_at DATETIME,
    expires_at DATETIME,
    created_by VARCHAR(20) NOT NULL DEFAULT 'payment',
    FOREIGN KEY (plan_id) REFERENCES plans (id)
);
```

```
CREATE TABLE vouchers (
    id INTEGER PRIMARY KEY,
    code VARCHAR(20) NOT NULL UNIQUE,
    plan_id INTEGER NOT NULL,
    redeemed_at DATETIME,
    expires_at DATETIME,
    FOREIGN KEY (plan_id) REFERENCES plans (id)
);
```

```
CREATE TABLE admins (
    id INTEGER PRIMARY KEY,
    username VARCHAR(40) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    created_at DATETIME NOT NULL
);
```

```
CREATE TABLE audit_log (
    id INTEGER PRIMARY KEY,
    admin_id INTEGER,
    action VARCHAR(60) NOT NULL,
    details TEXT,
    created_at DATETIME NOT NULL
);
```

Listing 6.1. Reference database schema.

6.3 Data protection and retention

- The system shall store only the data needed to run the service: phone number, device MAC address, plan, payment reference and timestamps.
- Mobile money PINs, card data and passwords of tenants shall never be handled or stored.
- Admin passwords shall be stored using PHP's password hashing functions.
- Records older than the retention period (OD-8) shall be deleted or anonymised.
- The owner shall handle tenant data in line with applicable Cameroonian personal-data legislation (to be verified), and a short privacy notice shall be shown on the help page.
- Database backups shall be taken regularly and stored away from the portal server.

7. Network and Router Configuration Requirements

This section lists what the router must be configured to do. It is deliberately written as requirements and not as command scripts, because exact settings depend on the router model and RouterOS version.

ID	Area	Requirement
NR-01	Starlink	The Starlink router shall be in bypass mode so the MikroTik is the only gateway for tenants. Use the Ethernet adapter where the dish model requires it.
NR-02	WAN and NAT	The MikroTik WAN shall obtain its address from Starlink by DHCP, with NAT masquerade for the LAN.
NR-03	LAN	All tenant Wi-Fi and wired ports shall belong to one bridge with its own DHCP pool. The portal server shall have a reserved address.
NR-04	Hotspot	The Hotspot shall run on the tenant bridge, with MAC-address login enabled in addition to username and password.
NR-05	User profiles	One profile per plan, each with a rate limit, one shared user and a connection limit.
NR-06	Expiry	Subscription expiry shall be enforced by the router, using the method chosen in OD-1 (for example an on-login expiry script with a scheduler). The router clock shall be synchronised by NTP.
NR-07	Walled garden	Unauthenticated devices shall reach only the portal server and, if hosted checkout is used, the listed Fapshi domains.
NR-08	Login page	The hotspot login page shall be customised with Buy a plan and voucher entry, and shall forward MAC and IP to the portal.
NR-09	TTL filter	A firewall rule on forwarded hotspot traffic shall drop packets whose TTL is not 64 or 128. IPv6 shall be covered or disabled.
NR-10	API access	A dedicated API user with only the permissions needed shall exist. The API service shall accept connections only from the portal server's IP.
NR-11	Management security	Default accounts shall be changed or disabled. Unused services shall be disabled. Router management shall not be reachable from tenant addresses.
NR-12	Backups	A router configuration export shall be saved after every change and kept off the router.
NR-13	Updates	RouterOS shall be updated in planned maintenance windows after reading release notes.

8. Non-Functional Requirements

8.1 Performance and capacity

ID	Requirement
NFR-PF-01	Portal pages shall load in under 3 seconds over the local Wi-Fi.
NFR-PF-02	Provisioning shall complete within 10 seconds of confirmed payment under normal conditions.
NFR-PF-03	The portal shall support at least 60 simultaneous tenant sessions and 100 registered tenants (design target, to be adjusted by the owner).
NFR-PF-04	Expired subscriptions shall lose access within 2 minutes of their expiry time.

Speed limits per plan should be chosen from the available Starlink bandwidth. A simple planning formula for the guaranteed share per active user is:

> **$B_{user} = (B_{total} \times u) / N_{active}$**
> where B_{total} is the measured Starlink throughput, u is a utilisation factor (for example 0.8, leaving headroom for overhead and bursts), and N_{active} is the number of tenants online at the same time. Example only: B_{total} = 100 Mbps, u = 0.8 and N_{active} = 40 gives B_{user} = 2 Mbps.

Starlink throughput varies with weather and obstructions, so B_{total} should be measured at different times of day before plan speeds are fixed.

8.2 Security

ID	Requirement
NFR-SC-01	Fapshi credentials and router passwords shall be stored in a configuration file outside the web root and never sent to the browser.
NFR-SC-02	All database access shall use prepared statements to prevent SQL injection.
NFR-SC-03	All output to pages shall be escaped to prevent cross-site scripting.
NFR-SC-04	Admin sessions shall use secure session settings, expire after inactivity, and be protected against cross-site request forgery.
NFR-SC-05	Login attempts to the admin portal and voucher redemption shall be rate limited.
NFR-SC-06	Webhook requests shall never be trusted on their own. Every payment shall be verified with Fapshi before access is granted.
NFR-SC-07	The router API shall be reachable only from the portal server and shall use an account with minimal permissions.
NFR-SC-08	The admin portal shall be reachable only from the local network or configured addresses, and shall use HTTPS where possible.
NFR-SC-09	Error messages shown to tenants shall not reveal internal paths, queries or credentials.

8.3 Reliability and availability

- The router and portal server shall start automatically after a power cut, with no manual steps.
- Existing subscriptions shall keep working on the router even when the portal server is off.
- A UPS or battery backup should keep the router and portal server running through short power cuts.
- The database shall be backed up automatically and the backup restored in a test at least once.
- Provisioning failures shall be retried automatically and shown on the dashboard, never lost silently.

8.4 Usability

- A tenant shall be able to buy a plan in no more than four taps and one phone number entry.
- All instructions shall be written in simple English and French.
- The interface shall work on small screens and low-end browsers without requiring installation.
- The admin portal shall let the owner approve a laptop in under one minute.

8.5 Maintainability and portability

- Code shall be organised into small PHP files with one clear responsibility each, and the Fapshi and router integrations shall each live in one module.
- Settings such as plan data, timeouts and addresses shall be in the database or a configuration file, not hard-coded in several places.
- Code shall be commented where behaviour is not obvious, with one statement per line for easy reading.
- The portal shall run on any Linux or Windows machine with PHP 8.x, a web server and SQLite or MySQL, so the server can be replaced easily.

8.6 Legal and commercial

- A short terms-of-use and acceptable-use notice shall be shown on the help page.
- A refund and dispute procedure shall be defined by the owner and described on the help page.
- Fapshi fees and any applicable taxes shall be considered when setting plan prices.

9. Use Cases

9.1 UC-01 Tenant buys a plan

Actor	Tenant
Preconditions	Device is connected to the Wi-Fi, has no active subscription, and the service is up.
Main flow	1. Login page appears. 2. Tenant taps Buy a plan. 3. Tenant selects a plan. 4. Tenant enters a mobile money number. 5. System starts the payment with Fapshi. 6. Tenant approves on the phone. 7. System confirms the payment with Fapshi. 8. System creates the MAC-bound hotspot user. 9. Device is logged in and the status page opens.
Alternative flows	A1. Payment declined or cancelled: system shows the failure and offers Retry. B1. Payment confirmed but router unreachable: tenant sees Payment received, activating shortly, and the system retries. C1. Service down: Buy a plan is disabled with a notice.
Postconditions	An active subscription exists for the device MAC address until the expiry time.

9.2 UC-02 Tenant checks status

Actor	Tenant
Preconditions	Device is connected.
Main flow	Tenant opens the status page. System identifies the device and shows plan, expiry and remaining time.
Alternative flows	A1. No active subscription: system offers Buy a plan. B1. MAC mismatch: system advises contacting the owner.
Postconditions	None.

9.3 UC-03 Owner approves a tenant's laptop

Actor	Owner
Preconditions	Tenant has paid for a plan and asks for laptop access. Laptop is connected to the Wi-Fi but not logged in.
Main flow	1. Owner logs in to the admin portal. 2. Opens Devices. 3. Selects the tenant's laptop from the list of unauthorised devices. 4. Chooses the tenant name and plan or extra-device option. 5. Confirms. 6. System creates a MAC-bound hotspot user and records the action in the audit log. 7. Laptop gains access.
Alternative flows	A1. Device not in the list: owner asks the tenant to reopen the browser so the router registers the host, then refreshes. B1. Router API error: system shows the error and logs it.
Postconditions	Laptop has an active subscription tied to its MAC address.

9.4 UC-04 Owner re-binds a subscription

Actor	Owner
Preconditions	A tenant's device MAC changed and the old subscription is still active.
Main flow	1. Owner opens Subscriptions and finds the tenant. 2. Selects the new device from the connected devices list. 3. Confirms re-bind. 4. System moves the hotspot user to the new MAC, keeping the original expiry time. 5. Action is logged.
Alternative flows	A1. Subscription already expired: system advises creating a new one.
Postconditions	Only the new device can use the subscription.

9.5 UC-05 Owner issues a cash voucher

Actor	Owner, Tenant
Preconditions	Tenant has paid the owner in cash.
Main flow	1. Owner generates a voucher for the plan. 2. Owner gives the code to the tenant. 3. Tenant enters the code on the login page. 4. System validates the code, binds it to the device MAC and creates the hotspot user.
Alternative flows	A1. Code already used or expired: system rejects it. B1. Too many failed attempts: system temporarily blocks further tries.
Postconditions	Voucher marked redeemed. Subscription active.

9.6 UC-06 Starlink outage

Actor	System
Preconditions	Internet connection or Fapshi becomes unreachable.
Main flow	1. Health check fails. 2. Portal shows service temporarily down and disables Buy a plan. 3. Existing subscriptions remain valid on the router. 4. When the connection returns, the system reconciles pending orders with Fapshi and provisions any that succeeded.
Alternative flows	A1. Owner issues cash vouchers or approves devices manually during the outage.
Postconditions	No paid tenant is left without access and no unpaid tenant is given access.

10. Technology Stack and Project Structure

Layer	Choice	Reason
Frontend	HTML, CSS, JavaScript (no framework)	About four pages. Fast on low-end phones. Easy to serve through the walled garden.
Backend	Raw PHP 8.x	Developer's existing skill. Small surface area, no framework overhead.
Web server	Apache or Nginx	Common, well documented, runs on Linux and Windows.
Database	SQLite (small deployments) or MySQL/MariaDB	SQLite needs no separate server. MySQL is an easy upgrade if needed.
Router	MikroTik RouterOS 7 Hotspot	Built-in captive portal, user profiles, expiry, speed limits and API.
Payments	Fapshi	Supports MTN Mobile Money and Orange Money in Cameroon.
Tunnel (optional)	Free tunnel service	Gives Fapshi a public HTTPS address for webhooks without opening ports.

Suggested folder structure

```
portal/
    public/
        index.html
        checkout.html
        status.html
        help.html
        assets/
            style.css
            app.js
        api/
            create_order.php
            order_status.php
            fapshi_webhook.php
            my_status.php
        admin/
            login.php
            dashboard.php
            devices.php
            subscriptions.php
            payments.php
            vouchers.php
            plans.php
            audit.php
    src/
        config.php
        db.php
        fapshi.php
        mikrotik.php
        subscriptions.php
        auth.php
    cron/
        reconcile.php
        expire.php
    storage/
        database.sqlite
        logs/
```

Listing 10.1. Proposed project layout. The src and storage folders sit outside the public web root.

11. Risk Analysis

ID	Risk	Impact	Likelihood	Mitigation
R-01	Tenants share their connection by tethering.	Revenue loss, congestion	High	TTL filter, per-user speed and connection limits, pricing of multi-device plans, honest documentation of limits.
R-02	Phone MAC address changes and the tenant loses access.	Support effort	Medium	Clear message on the portal, admin re-bind, help page instructions.
R-03	Power cuts stop the router and portal server.	Service interruption	High	UPS or battery backup, automatic start, router-side enforcement.
R-04	Starlink outage during a payment.	Tenant paid without access	Medium	Reconciliation job, outage notice, manual approval by the owner.
R-05	Fapshi API change or downtime.	Payments fail	Low to medium	Isolated integration module, monitoring, vouchers as fallback.
R-06	Forged or repeated webhook calls.	Free access, duplicate credit	Medium	Verify every transaction with Fapshi, unique transaction identifiers.
R-07	Router API credentials leaked.	Full control of access	Low	Dedicated low-privilege account, IP restriction, secrets outside web root.
R-08	MAC cloning to use another tenant's subscription.	Account abuse	Low	One session per user limits impact. Audit log helps the owner investigate.
R-09	Captive-portal pop-up does not appear on some phones.	Tenant confusion	Medium	Help page with manual address, test across common phones before launch.
R-10	Private DNS or VPN on tenant phone blocks the redirect.	Cannot reach the portal	Medium	Help guidance to switch these off while signing in.
R-11	Single server failure.	No new sales	Medium	Regular backups, simple reinstall guide, replaceable hardware.

12. Testing and Acceptance Criteria

12.1 Test approach

- **Unit and module tests:** PHP functions for order creation, status handling, duplicate protection and router calls, using a mocked Fapshi and a test router profile.
- **Integration tests:** full payment flow in Fapshi's sandbox or with small real amounts, and router provisioning on a real MikroTik.
- **Device tests:** at least Android (two brands), iPhone, Windows laptop and macOS laptop.
- **Failure tests:** payment declined, webhook repeated, router unreachable, Starlink unplugged, server power cycle.
- **Security tests:** attempts to reach the internet before payment, to reuse a code on a second device, to tether, to access admin pages from a tenant device, and to inject input into forms.

12.2 Acceptance criteria

ID	Criterion	Related requirements
AC-01	An unpaid device cannot reach any external website or app, and can open only the portal.	FR-CP-01, FR-CP-04
AC-02	A tenant can buy a plan on a phone and be online within 10 seconds of confirming payment.	FR-PY, FR-AP-01 to FR-AP-03
AC-03	A second device cannot use a subscription bought for the first device.	FR-DB-01 to FR-DB-03
AC-04	A device sharing its connection by Wi-Fi hotspot or USB tethering is blocked in standard tests.	FR-AT-01
AC-05	A repeated or forged payment notification creates no second subscription and grants no access.	FR-PY-04, FR-PY-06
AC-06	An expired subscription loses access within 2 minutes.	FR-AP-05, NFR-PF-04
AC-07	The owner can approve a connected laptop in under one minute and revoke it immediately.	FR-AD-02 to FR-AD-04
AC-08	With the portal server switched off, existing subscribers stay online until expiry.	FR-OU-05
AC-09	With Starlink unplugged, the portal shows the outage notice and disables payments. After reconnection, a payment started earlier is reconciled.	FR-OU-01 to FR-OU-03
AC-10	Fapshi credentials do not appear in any page source, network response or log.	NFR-SC-01
AC-11	The admin portal cannot be opened from a tenant's device.	FR-AD-11
AC-12	A cash voucher works once, binds to one device and cannot be reused.	FR-VC-01, FR-VC-02

13. Deployment and Operations

13.1 Deployment steps

1. Install and test the Starlink dish, then enable bypass mode on its router.
2. Configure the MikroTik: WAN, LAN bridge, DHCP, NAT, time synchronisation and firewall basics.
3. Set up the Hotspot with MAC login, plan profiles, expiry method, walled garden and the TTL filter.
4. Install the web server, PHP and database on the portal server and give it a reserved LAN address.
5. Deploy the portal files, configuration file (Fapshi and router credentials) and database schema.

6 Create the router API account and restrict it to the portal server.
7 Customise the hotspot login page and point it to the portal.
8 Test with several phones and laptops using the acceptance criteria.
9 Set up automatic backups, scheduled reconciliation and expiry jobs.
10 Train the owner on the admin portal, then open the service to tenants.

13.2 Ongoing operations

Task	Frequency
Check the dashboard for stuck payments and provisioning errors	Daily
Reconcile Fapshi transactions with subscriptions	Daily (automatic) and weekly (manual review)
Back up the database and router configuration	Daily database, after each router change
Review the audit log	Weekly
Measure Starlink throughput and review plan speeds	Monthly
Update PHP, web server and RouterOS	Planned maintenance, after reading release notes
Test restoring a backup	Every three months

14. Future Enhancements

- Tenant self-service extra-device add-on with its own price.
- SMS or WhatsApp reminders before a subscription expires.
- Data-volume plans and a daily fair-use cap.
- Revenue and usage charts for the owner, with monthly reports.
- Support for several routers or buildings from one admin portal.
- Referral discounts and promotional codes.
- Automatic renewal prompts at expiry.

15. Appendix A: Traceability Summary

The table links each business objective to the requirement groups that deliver it, so reviewers can check that every objective is covered.

Business objective	Requirement groups
Tenants can pay by mobile money without owner involvement	FR-CP, FR-PL, FR-PY, FR-AP, NFR-PF-01, NFR-PF-02
Access lasts for exactly the period paid for	FR-AP-05, FR-AP-06, NR-06, NFR-PF-04
One payment covers one device only	FR-DB, FR-AT, NR-04, NR-05, NR-09
Owner can serve special cases such as laptops and cash payments	FR-AD, FR-VC, UC-03 to UC-05
Payments are trustworthy and cannot be faked or duplicated	FR-PY-04, FR-PY-06, NFR-SC-06, AC-05
Service degrades safely during outages and power cuts	FR-OU, NFR reliability, R-03, R-04, AC-08, AC-09
System is simple to run, low cost and easy to maintain	Design constraints, section 10, section 13
Tenant and payment data is handled responsibly	Section 6.3, NFR-SC, FR-LG

16. Appendix B: Items to Confirm Before Development

- Plan names, durations, speeds and prices (OD-2).
- Subscription clock rule (OD-1).
- Final hosting choice for the portal server (OD-3).
- Fapshi payment method (hosted checkout or direct charge) and exact API details (OD-4, OD-5).
- Router model, RouterOS version and Starlink dish model (OD-7).
- Data retention period and wording of the privacy and terms notices (OD-8).
- Whether English only or English and French interfaces are required at launch (OD-9).